

Nicholas (Nick) Wagner

email: nicholas_wagner@brown.edu

website: nlwagner.github.io

Education

Ph.D. 2024

Dissertation: Deformation of the Martian Lithosphere Through Time

Baylor University

Advisor: Dr. Peter James

B.Sc. 2019

Major: Geophysics

Minor: Computer Science Specializing in Data Science

Colorado School of Mines

Work Experience

2024–Present

Postdoctoral Research Associate

Brown University

2019–2024

Research & Teaching Assistant

Baylor University

2018–2019

Data Processing Technician

EDCON-PRJ

Research Experience

2023

Telecomms Chair, Gravity Science Lead

NASA/JPL Planetary Science Summer School

“New Frontiers Class Titan Orbiter Concept”

2022

Student Observer of STM25

InSight

“InSightSeer Program”

2021

Study Participant

Keck Institute for Space Studies Workshop

“Next-Generation Planetary Geodesy”

2021

Research Intern

NASA/JPL

2019

Student Contractor

United States Geological Survey

2018

Research Intern

Lunar and Planetary Institute

2018 – 2019

Undergraduate Research Fellow

Colorado School of Mines

Publications

In Review

- **Wagner, N.**, Lau, H., Berne, A., Frankenberg, N., Matsuyama, I. Geodetic Evidence for a Compositionally Distinct Lunar Nearside Mantle
- Manuscript is currently in review at *Nature*.

2026

- Berne, A., **Wagner, N.**, Matsuyama, I., Goossens, S., Cheng, K., Genova, A., Riviera-Navarro M., Bagheri, A., Qin, C., Marusiak, A., Hemingway, D., Lau, H., Zhong, S., Nimmo, F. Molten Root for Mars's Crustal Dichotomy Inferred from Seasonal Tidal Tomography *Nature* <https://doi.org/10.1038/s41586-026-10893-x>
- Project led by Alex Berne. I contributed expertise on polar cap and atmospheric effects to the time-variable gravity signal of Mars. I also contributed to the interpretation of our results in terms of the geodynamic state of Mars.
- Broad, K., Sadler, B., James, P., Hoover, S., **Wagner, N.**, Hood, D. A geophysical survey of the Kentland Crater formation *MDPI Geosciences* <https://doi.org/10.3390/geosciences16040155>
- Project led by Katie Broad. I helped provide field work for collecting magnetotelluric data.

2025

- **Wagner, N.** & James, P. New Geophysical Constraints for Intrusive Magmatism at Large Martian Volcanoes: Implications for Crustal Thickness and Volatile Outgassing *Journal of Geophysical Research: Planets* <https://doi.org/10.1029/2025JE008959>

2024

- Seltzer, C., Bott, K., Brouwer, G., Burt, D., Gentgen, C., Lien, R., Abbate, J., Head, T., Hill, J., Gandarillas, V., Green, A., Larson, J., Montiel, N., Moreno, R., Mullikin, E., Radzom, B., **Wagner, N.**, Wijesekara, P., Nash, A., Keane, J. THUNDER: A Titan Orbiter New Frontiers Mission Concept *Planetary Science Journal*. <https://doi.org/10.3847/PSJ/ad973e> - I contributed expertise related to measuring the gravity field of Titan, and it would tell us about the state of formation of the moon.
- **Wagner, N.**, James, P., Ermakov, A., Sori, M. Evaluating the Use of Seasonal Surface Displacements and Time-Variable Gravity to Constrain the Interior of Mars *Journal of Geophysical Research: Planets*. <https://doi.org/10.1029/2023JE008053>

2022

- Xiao, H., Stark, A., Schmidt, F., Hao, J., Steinbrügge, G., **Wagner, N.**, Su, S., Cheng, Y., Oberst, J. Spatio-temporal level variations of the Martian Seasonal North Polar Cap from co-registration of MOLA profiles *Journal of Geophysical Research: Planets*. <https://doi.org/10.1029/2021JE007158>
- I contributed expertise in regards to how measured variations in the polar caps of Mars can induce periodic deformations of the surface.

In Prep

- **Wagner, N.**, James, P., McGovern, P. The Argument for a Geodynamically Controlled Emplacement of Pavonis Fossae
- Manuscript is written, but some analysis will need to be redone. Estimated time of submission: within 6 months.
- Jindal, A., **Wagner, N.**, Trussell, A., O'Rourke, J., Head, J., Kerr, M., Hayes, A., Birch, S. Investigating Global Patterns of Tectonic Deformation in Venusian Canals
- Project led by Abhinav Jindal. I am contributing expertise on lithospheric flexure and stress modeling and contributing to the discussion. Analysis is finished, and the manuscript is currently being written. Estimated time of submission: within 2 months.

- North, A., James, P., **Wagner, N.**, Moriarty, D. Geophysical Constraints on Bulk Density at the Gruithuisen Domes
- Project led by Allie North. I am contributing expertise on gravity modeling and Lunar volcanism. Manuscript is currently in revisions amongst coauthors. Estimated time of submission: within 3 months.

Awards and Fellowships

LunaSCOPE Postdoctoral Research Fellow <i>2024–Present</i>	Brown University
Graduate School Travel Award <i>2020–2024</i>	Baylor Graduate School
Graduate School Fellow <i>2019–2024</i>	Baylor Graduate School Fellowship
Outstanding Teaching Assistant <i>2022</i>	Baylor Geosciences Department
Texas Space Grant Fellow <i>2020</i>	Texas Space Grant Consortium

Teaching Experience

2026 <i>Instructor</i>	Brown University
CEES0959: Strange New Worlds: A Crash Course in Planetary Mission Design	
2019–2024 <i>Teaching Assistant</i>	Baylor University
GEO1401: Earthquakes and Natural Hazards	
GEO3319: Intro to Geophysics	
GEO5328: Geodynamics	
2019 <i>Teaching Assistant</i>	Colorado School of Mines
GPGN268: Geophysical Data Analysis	

Conference Abstracts ([†]indicates oral presentation)

2025

- [†]**Wagner, N.**, Lau, H., Berne, A., Frankenberg, N. *The Lopsided Moon: Tidal Signals of a Heterogeneous Interior* (AGU)
- Frankenberg, N., **Wagner, N.**, Lau, H. *The geochemical context of geophysically derived nearside-farside Lunar mantle variations* (AGU)
- Wieczorek, M., Alvarez-Candal, M., Angrisani, M., Attree, N., Bhattacharya, A., Blewett, D., Broquet, A., Burkhard, L., Cascioli, G., Collins, G., Cone, K., Crameri, F., Denton, A., Elkins-Tanton, L., Fa, W., Filiberto, J., Flahaut, J., Frigeri, A., Guimond, C., Guray Hatipo, Y., Jacobson, S., Kim, D., Lucas, A., Maia, J., Mandon, L., Martinot, M., Miljkovic, K., Montesi, L., Nawal, A., Oran, R., Pan,

L., Park, R., Putzig, N., Rodriguez, S., Roos-Serote, M., Root, B., Ruedas, T., Schmidt, F., Snape, J., Tai Udovicic, C., Tosi, N., van der Wal, W., **Wagner, N.** *Planetary Research - A Diamond Open Access Journal For Planetary Science* (LPSC)

- **Wagner, N.**, James, P. *Geophysical Evidence for High I/E Ratios at Large Martian Volcanoes* (LPSC)

2024

- **Wagner, N.**, James, P., Ermakov, A., Sori, M. *Using Seasonal Surface Displacements and Time-Variable Gravity to Constrain the Interior of Mars* (Mars Geophysics After InSight)

- North, A., James, P., **Wagner, N.**, Moriarty, D., Kiefer, W., Siegler, M. *Crustal Context of the Moon's Gruithuisen Domes* (LPSC)

- [†]**Wagner, N.**, James, P. *Higher Intrusive Magma Volumes Found at Olympus Mons from Gravity and Topography Data* (LPSC)

2023

- **Wagner, N.**, James, P., Ermakov, A., Sori, M. *New Estimates of Seasonal Surface Displacements and Time Variable Gravity on Mars* (LPSC)

- North, A., James, P., **Wagner, N.**, Moriarty, D. *Gravity Constraints on Bulk Density and Igneous Intrusion at Gruithuisen Domes* (LPSC)

- Moriarty, D., James, P., North, A., **Wagner, N.** *Evidence for Widespread Silicic Lithologies Across the Gruithuisen-Mairan Region* (LPSC)

- Morris, J., Schurmeier, L., **Wagner, N.**, Baker, S., Bill, C., Mohanna, S., Roth, N., Sanderson, H., Sridhar, S., Woodley, S., Daubar, I., Fernando, B., Newman, C., Panning, M. *InSightSeers: Peering into Invited Student Participation of a Mission Science Team Meeting* (LPSC)

2022

- **Wagner, N.**, James, P., Ermakov, A., Sori, M. *Quantifying Lithospheric Deflection Caused by Seasonal Mass Transport from the Polar Layered Deposits on Mars* (LPSC)

- **Wagner, N.**, Park, R., James, P. *A Geophysical Investigation of the Compensation State of Hellas Planitia* (LPSC)

- Sori, M., Bramson, A., Byrne, S., James, P., Ojha, L., **Wagner, N.** *Gravity Science Constrains the Presence and Volume of Mid-Latitude Ice Sheets on Mars* (LPSC)

- Keane, J., Sori, M., Ermakov, A., Austin, A., Bapst, J., Berne, A., Bierson, C., Bills, B., Boening, C., Bramson, A., D'Amico, S., Denton, A., Evans, A., Hemingway, D., Hernandez, S., Hogstrom, K., Izquierdo, K., James, P., Johnson, B., Kahre, M., Lau, H., Navarro, T., Neveu, M., Nimmo, F., O'Rourke, J., Ojha, L., Paik, H.J., Park, R., Rosen, P., Simons, M., Smith, D., Smrekar, S., Soderlund, K., Steinbrügge, G., Tikoo, S., Vance, S., **Wagner, N.**, Weber, R., Zebker, H., Zuber, M. *Next-Generation Planetary Geodesy: Results from the 2021 Keck Institute for Space Studies Workshops* (LPSC)

- Sori, M. M., Ermakov, A. I., Keane, J. T., Bierson, C. J., Bills, B. G., Bramson, A. M., D'Amico, S., Evans, A. J., Hemingway, D. J., Izquierdo, K., James, P. B., Johnson, B. C., Kahre, M. A., Navarro, T., O'Rourke, J. G., Ojha, L., Paik, H. J., Park, R. S., Simons, M., Smith, D. E., Smrekar, S. E., Soderlund, K. M., Steinbrügge, G., Tikoo, S. M., Vance, S. D., **Wagner, N. L.**, Weber, R. C., Zebker, H. A. *Next-Generation Planetary Geodesy: Results from the 2021 Keck Institute for Space Studies Workshops* (Low-Cost Science Mission Concepts for Mars Exploration)

2021

- **Wagner, N.**, James, P. *Evaluating Magma Ascent at Pavonis Mons, Mars Using Stress from Flexure* (LPSC)

2019

- **Wagner, N.**, Kay, J., Schenk, P. The Orientation of the Bladed Terrain Feature in Tarturus Dorsa, Pluto and Possible Reorientation of Pluto (LPSC)

2018

- **Wagner, N.**, Kay, J., Schenk, P. Study of the Orientation of the Bladed Terrain Feature in Tartarus Dorsa, Pluto (AGU)

Mentorship

Neal Frankenberg <i>Thermodynamic and geochemical interior modeling of the lunar interior: a geophysical perspective</i>	Brown Undergraduate <i>2025-2026</i>
Allie North <i>Gravity survey of silicic volcanism on the Moon</i>	Baylor Undergraduate <i>2022-2024</i>
Skylar Hoover <i>Magnetotelluric survey of Kentland Crater, Indiana</i>	Baylor Undergraduate <i>2022-2023</i>

Outreach & Seminars

RI Public Schools <i>Judge for State High School Science Fair</i>	April 2026 <i>Providence, RI</i>
Geophysical Fluid Dynamics Group <i>Geodetic Evidence for a Compositionally Distinct Lunar Nearside Mantle</i>	May 2026 <i>ETH Zurich</i>
AstroAssembly <i>Is the Moon Made of Parmesan or Brie?</i> <i>How Tides Tell Us What's Inside the Moon</i>	October 2025 <i>Seagrave Observatory</i>
Astronomy on Tap <i>Is the Moon Made of Parmesan or Brie?</i> <i>How Tides Tell Us What's Inside the Moon</i>	June 2025 <i>Moniker Brewing</i>
Lunch Bunch <i>Tidal Tomographic Insights into Lunar Interior Structure:</i> <i>Is the Moon Made of Parmesan or Brie? the Moon</i>	April 2025 <i>Brown University</i>
Solid-Earth Seminar <i>Martian Geophysics and Geodesy:</i> <i>Insights into Volcanism and Planetary Structure</i>	October 2024 <i>Georgia Tech</i>
Special Geodesy Seminar <i>Next Generation Martian Geodesy: What Could The Seasonal Polar Caps Tell Us About The Martian Interior?</i>	July 2024 <i>NASA Goddard</i>

Organizations and Competitions

2025–2026
Brown University

Planetary Lunch Bunch Organizer

2020–2023
Geoscience Representative (Baylor University)

Graduate Student Association (GSA)

2017–2019
Senate Representative (Colorado School of Mines)

Undergraduate Student Government (USG)

2017–2018
Deputy Project Lead and Quality Control Lead

2018 NASA Rasc-AI Competition

Presented at AIAA 2018: Paper number AIAA-2018-5108. <https://doi.org/10.2514/6.2018-5108>

2018
Final Report Lead

Colorado School of Mines Geophysics Summer Field Camp

Led the completion of a 200-page report detailing the findings of the field camp's surveys in Pagosa Springs, CO.

2018
Treasurer

Student Society of Geophysicists (SSG)

Scientific Service

2025
Participant

NASA Artemis IV LSSW

2021–2026
Executive Secretary and Panelist

NASA Grant Review Panels
ROSES, PRISM

2025
Journal Reviewer

Science: Advances

2025
Journal Reviewer

AGU: Geophysical Research Letters

2022–2026
Journal Reviewer

AGU: JGR-Planets

2023–2025
Journal Reviewer

Elsevier: Icarus